

TYPES OF RAG

Naive RAG (Basic RAG)

Naive RAG is the simplest implementation of retrieval-augmented generation, where the system retrieves the top relevant documents from a vector database and directly feeds them into the language model without any filtering or optimization. It follows a straightforward pipeline: user query → embedding → retrieval → LLM response. However, since there is no ranking or refinement, irrelevant or noisy data may be included, which can reduce accuracy.

For example, a basic college FAQ chatbot that retrieves multiple documents about exam schedules and directly generates an answer from them is a typical use case of naive RAG.

Advanced RAG

Advanced RAG enhances the basic RAG pipeline by improving the quality of retrieved data before sending it to the language model. It uses techniques such as query rewriting, hybrid search (combining keyword and semantic search), re-ranking of documents, and context filtering to ensure only the most relevant information is passed. This significantly improves accuracy and reduces hallucination.

For example, a legal AI assistant that refines the user query, retrieves multiple law documents, ranks them based on relevance, and selects only the most important sections to generate a precise answer is an example of advanced RAG.

Modular RAG

Modular RAG structures the RAG system into independent components such as retriever, ranker, generator, and memory/cache, allowing each module to be developed, scaled, and optimized separately. This architecture is commonly used in large-scale production systems where flexibility and maintainability are important. Instead of a single pipeline, each module can be replaced or upgraded independently.

For example, an enterprise AI platform where the retrieval system runs as one service, the ranking model as another, and the LLM as a separate API is a real-world example of modular RAG.

Hybrid RAG

Hybrid RAG combines multiple retrieval techniques, typically semantic search (vector-based) and keyword-based search (like BM25), to improve the relevance of retrieved documents. Semantic search helps understand the meaning of the query, while keyword search ensures exact matches are not missed. By combining both, the system achieves better recall and precision.

For example, an e-commerce search system where a user searches for “iPhone 13 battery issue” will benefit from both exact keyword matching (“battery issue”) and semantic understanding of the problem, making hybrid RAG very effective.

Graph RAG

Graph RAG uses knowledge graphs instead of traditional document-based retrieval, where data is stored as entities (nodes) and relationships (edges). This allows the system to perform reasoning based on relationships between data points rather than just text similarity. It is particularly useful for complex queries involving connections between entities.

For example, in an organization, answering a query like “Who manages the employee working on Project X?” requires traversing relationships between employee, project, and manager, which is efficiently handled by graph RAG.

Agentic RAG

Agentic RAG introduces intelligent agents that dynamically decide how to retrieve and process information instead of following a fixed pipeline. The agent can plan steps, refine queries, call external tools or APIs, and perform multi-step reasoning before generating the final answer. This makes it suitable for complex and multi-step queries.

For example, an AI research assistant that gathers data from multiple sources, compares results, and then generates a summarized explanation demonstrates agentic RAG.

Multimodal RAG

Multimodal RAG extends traditional RAG by supporting multiple data types such as text, images, audio, and video. It converts all these formats into embeddings and retrieves relevant information across different modalities before generating a response using a multimodal language model. This enables richer understanding and more comprehensive answers.

For example, a medical AI system that analyzes an X-ray image along with a patient’s report and provides a diagnosis is a practical use case of multimodal RAG.

Adaptive RAG

Adaptive RAG is a dynamic approach where the system decides whether retrieval is needed based on the user query. Instead of always performing retrieval, it uses a query classifier or decision mechanism to determine if the question can be answered directly by the LLM or requires external data. This reduces cost, latency, and unnecessary processing.

For example, if a user asks “What is machine learning?”, the system answers directly using the LLM, but if the question is “What is my company’s leave policy?”, it triggers retrieval to fetch internal documents, making adaptive RAG highly efficient.

WCAG

In countries like United States, United Kingdom, India, and across the European Union, the **government and public sector** are the major users of WCAG. It is often legally mandatory for government websites such as tax portals, identity systems, and public service platforms to follow accessibility guidelines so that all citizens, including people with disabilities, can access essential services without barriers.

In the **banking and financial services industry**, organizations use WCAG to ensure that digital banking platforms, mobile applications, and payment systems are accessible. This helps users perform tasks like checking account balances, transferring money, and managing financial data using assistive technologies such as screen readers and keyboard navigation.

In the **healthcare industry**, WCAG is used to make patient portals, telemedicine platforms, and medical information systems accessible. Hospitals and healthcare providers ensure that patients can book appointments, access medical records, and understand health information easily, regardless of their physical or cognitive abilities.

In the **education sector**, including universities and online learning platforms, WCAG plays a key role in providing inclusive learning experiences. Educational content such as videos, PDFs, and online assessments are designed with captions, proper structure, and screen-reader compatibility so that all students can learn effectively.

In the **e-commerce and retail industry**, companies use WCAG to improve accessibility in product browsing, search, and checkout processes. This ensures that users with disabilities can independently shop online, which also helps businesses expand their customer base and improve user experience.

In the **technology and software industry**, WCAG is applied in websites, applications, and enterprise tools to ensure accessibility across digital products. Features like voice support, accessible UI components, and compatibility with assistive technologies help make software usable for a wider audience.

In the **media and entertainment industry**, WCAG is used to make digital content such as videos, streaming platforms, and news websites accessible. This includes providing captions, subtitles, audio descriptions, and accessible media players so that all users can consume content easily.